Supplementary Material: When MRI Slice Interpolators Disagree

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1 Reproducibility Scope

All primary results use complete evaluation profiles: 115 IXI examinations, 24 fastMRI-brain examinations, and 22 held-out NingXia examinations pooled across eight cross-validation folds. Predictions are matched by case identifier and checked for equal volume shape before scoring. The factor-four protocol evaluates slice indices not divisible by four. No reconstruction model is trained as part of the committee analysis.

For target model m, the all-member committee contains linear and cubic interpolation plus every available neural output except m. The neural committee removes linear and cubic; the analytical committee retains only those two references. The main voxel score is the median absolute target-to-member difference. Examination scores average this map inside a mask derived from the target prediction at 1% of its maximum absolute intensity.

2 Bootstrap Intervals

Table 1 reports 95% intervals from 2,000 case-bootstrap replicates for the four primary endpoints. Resampling is paired within a target–dataset result. For nAURC and RRR, the complete ranking and its random and oracle references are recomputed in every replicate. All foreground pixel-correlation, case-Spearman, and RRR_{50} intervals exclude zero, although the small-cohort intervals remain wide and should not be interpreted as evidence of a stable deployment threshold.

3 Committee Composition

Figure 1 gives target-level nAURC rather than only the dataset averages reported in the main text. Mean all/neural/analytical nAURC is 0.032/0.033/0.059 on IXI, 0.073/0.017/0.118 on fastMRI, and 0.058/0.094/0.031 on NingXia. Thus, neural peers are strongest on fastMRI and analytical references on NingXia. This reversal is evidence against choosing one committee composition after observing test errors; the all-member committee is retained as the pre-specified main analysis.

4 Leave-One-Member-Out Analysis

For each target and examination, we first average the target-to-member absolute difference inside the deployment-compatible foreground mask. We then remove one member and take the median of the remaining member scores. This score-level analysis is intentionally distinguished from the voxel-level main method: it tests whether examination ranking depends critically on one committee member without claiming an identical spatial map after removal. Complete results are provided in the machine-readable composition and leave-one-out report CSVs.

5 Proxy Comparisons

The all-member target-specific score is compared with neural-only disagreement, analytical-only disagreement, shared committee dispersion, and residuals to linear and cubic interpolation. Pixel-correlation differences use paired case bootstrap resampling. Ranking differences recompute nAURC after paired case resampling. Relative to shared dispersion, target-specific disagreement lowers nAURC by 0.011–0.025 for all four IXI targets with intervals excluding zero; differences on fastMRI include zero, and NingXia is mixed. Complete estimates and intervals are stored in `paired_proxy_bootstrap.csv`; retaining the machine-readable table avoids an unreadable multi-page matrix in the manuscript.

TTA variance is treated as an auxiliary source of sensitivity rather than a direct substitute for committee disagreement. It requires repeated inference and measures transformation response, whereas the proposed score measures inconsistency across reconstructors. Full-profile TTA, calibration-bin, failure-ranking, and reconstruction-context results are retained as machine-readable supplementary assets. They are not used to claim superiority for the corrected target-specific score because the earlier baseline-disagreement analysis used a shared committee map.

Table 1: Primary point estimates with case-bootstrap 95% intervals. RRR_{50} is expressed as a percentage.

Dataset	Target	n	Pixel r	Case ρ	nAURC / RRR_{50}
IXI	ACSI	115	0.435 [0.428,0.441]	0.954 [0.923,0.972]	0.035 [0.018,0.060] / 50.1 [46.1,54.8]%
IXI	HAT	115	0.511 [0.503,0.519]	0.966 [0.940,0.980]	0.024 [0.012,0.042] / 49.6 [45.4,53.9]%
IXI	I^3Net	115	0.351 [0.346,0.357]	0.951 [0.922,0.969]	0.044 [0.024,0.073] / 52.1 [45.9,55.9]%
IXI	SwinIR	115	0.519 [0.511,0.527]	0.965 [0.939,0.980]	0.025 [0.012,0.043] / 49.5 [45.1,53.6]%
fastMRI	HAT	24	0.325 [0.293,0.360]	0.916 [0.761,0.962]	0.067 [0.017,0.176] / 19.7 [11.7,26.1]%
fastMRI	I^3Net	24	0.289 [0.260,0.321]	0.890 [0.721,0.962]	0.092 [0.030,0.187] / 19.3 [10.8,25.8]%
fastMRI	SwinIR	24	0.359 [0.325,0.395]	0.931 [0.780,0.978]	0.061 [0.013,0.158] / 20.5 [12.6,26.9]%
NingXia	HAT	22	0.277 [0.256,0.296]	0.929 [0.786,0.974]	0.076 [0.016,0.239] / 22.2 [12.8,31.3]%
NingXia	I^3Net	22	0.355 [0.331,0.376]	0.914 [0.774,0.968]	0.063 [0.016,0.189] / 23.1 [12.6,32.7]%
NingXia	SwinIR	22	0.279 [0.256,0.300]	0.936 [0.804,0.985]	0.036 [0.007,0.120] / 21.4 [12.2,31.2]%

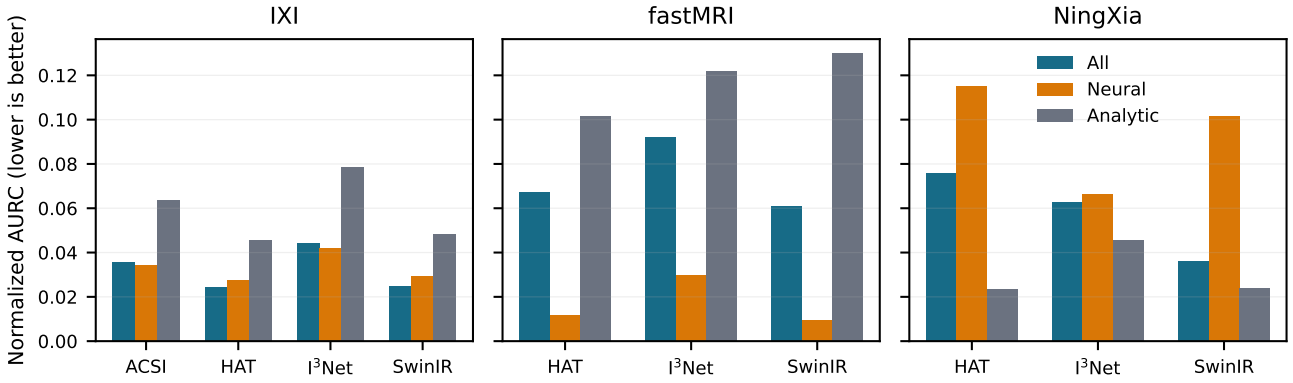


Figure 1: Target-level committee-composition sensitivity. Lower normalized AURC indicates a ranking closer to the oracle. Differences across targets expose correlated errors and dependence on analytical references.

6 Failure Detection and Interpretation

Binary failures are defined retrospectively as the highest-MAE 10% of examinations for each target model. This gives 12 positives on IXI and only three positives on each smaller cohort. AUROC, AUPRC, and top-k recall are therefore supplementary endpoints. The risk–coverage curve remains primary because it uses continuous reconstruction error and directly represents review workload.

High disagreement can be correct when a target preserves structure smoothed by other methods. Low disagreement can be wrong when committee members share a bias. The qualitative selection in the main paper includes both a useful referral and a false-reassurance example to make this boundary visible.

7 Artifact Traceability

The analysis writes per-case scores, per-member scores, summary metrics, risk curves, and failure tables into profile-specific directories. Statistical reports consume only those CSV files. Paper tables and figures consume only the report and profile CSVs and write an `asset_source_manifest.csv`. A final consistency check verifies required artifacts, formatted main-table values, forbidden screening-profile claims, and unresolved LaTeX references.